

Latin America URTeC: 3969226

Anomaly Detector Tool – Integrating Traditional Casing Collar Locator with Innovative Algorithm for Early Casing Anomaly Identification

Lucrecia Gava^{*1}, Lucas Goñi¹, Pablo Nache¹, Juan Carlos Bonapace¹ 1. Tecpetrol S.A.

Copyright 2023, Latin America Unconventional Resources Technology Conference (LA URTeC) DOI 10.15530/urtec-2023-3969226

This paper was prepared for presentation at the Latin America Unconventional Resources Technology Conference held in Buenos Aires, Argentina, 4-6 December 2023.

The LA URTeC Technical Program Committee accepted this presentation on the basis of information contained in an abstract submitted by the author(s). The contents of this paper have not been reviewed by LA URTeC and LA URTeC does not warrant the accuracy, reliability, or timeliness of any information herein. All information is the responsibility of, and, is subject to corrections by the author(s). Any person or entity that relies on any information obtained from this paper does so at their own risk. The information herein does not necessarily reflect any position of LA URTeC. Any reproduction, distribution, or storage of any part of this paper by anyone other than the author without the written consent of LA URTeC is prohibited.

Abstract

In the Vaca Muerta wells, operators face daily difficulties due to casing issues. The industry presently lacks efficient early-stage detection approaches for addressing these problems, which often results in decision-making only after the event has occurred. This situation leads to further operational complications and adverse economic consequences. In our research, we present a novel tool designed to detect anomalies, which will enable us to establish a contingency plan for effective response.

Unusual casing collar locator (CCL) readings sometimes indicate where the wireline gets stuck in later runs. Although not specifically designed for detecting casing issues, irregular CCL fluctuations might hint at problems with casing diameters. Initially, we applied an isolation forest-based anomaly detection algorithm to identify CCL disturbances, but not in real-time. In a recent pad, pump-down runs could not be executed due to casing restrictions. This incident led us to shift the frac fleet and use a multi-finger caliper tool. The results allowed us to tune the anomaly detector tool (ADT). Blind tests on historical stages showed our ADT successfully identified problematic stages from casing restrictions. The research seeks to use this tool as a first step to make informed decisions on contingency plans to mitigate related problems to this.

Introduction

Casing deformation has been observed in conventional and unconventional reservoirs, at the recent years has gained more attention in shale developments, several cases has been reported in Canada, China and Argentina. It is recommended to review two technical publications to have a better understanding about the possible causes of casing deformation generation. The first one, Casero and Rylance (2020) presented an initial approach about casing deformation that can be influenced by: a- well construction (tubular, cement), b-subsurface (tectonics and rock properties) and c-operations; and several case histories around the world. The second one, Russel (2021) shown different diagnostic tool to collect information about deformation; mechanisms and root causes in unconventional well focusing on cases with slip on natural fractures and bedding planes.

In this work, we will refer most of the time as casing restriction issues (CR) because we really know that we have a restriction to pass a particular well depth with the standard diameter of plug and perf bottom hole assembly. Only when we ran a specific diagnostic tool to obtain information defining the nature of the problem, we will refer as casing deformation (CD).

Vaca Muerta Experiences

From the beginning of the exploration phase in the Vaca Muerta (VM) formation, there has been a detection of casing restrictions in vertical wells. As the development of fields with multi-horizontal wells has progressed over the years, this issue has gained greater attention. Multiple operators have encountered this problem more than once, which has had a negative impact on the project's economy. An initial search of public information for this problem in VM has been carried out. The following are the main observation cited

- Garcia et al. (2013) shows two clear examples of casing deformation in vertical wells (Figure 1a)
- Rimedio et al. (2015) presented examples in Loma Campana field and mention the negatively impact in well performance.
- Gandomkar et al. (2021) reported an example of 33% of horizontal section loss during the completion of 3 horizontal wells at the same location, Multifinger Imaging Tool reveled the minimum internal diameter and final CT frac mill-out describe sever restrictions not being possible to reach the toe of the well.
- Maria et al. (2022) mention in La Calera field similar situations during plug & perf operation, the mitigation adopted for this scenario has been increase the stage length (double or triple) applying diverter to move away of these problem zone.
- Chao et al. (2022) documented in a well with casing restriction, the very good correlation between the section of the well without stimulation (loss stages) and the absence of 4D seismic signal, (volume rock stimulated along the lateral versus zone bypassed for this issues), (Figure 1b)
- Bonapace et al. (2023) identified casing restriction at different phases of the project (vertical and horizontal wells) impacting in drain lost (from 30 to 450m) with an extreme case of one well lost completely. Additionally, refers about the problem related to partial coiled tubing well clean out affecting well productivity.
- Mirande et al. (2023) reported the use of a simplified BHA for CT cleanout (special jet nozzle 73mm) in which reached the final well depth after passed 3 casing restriction.

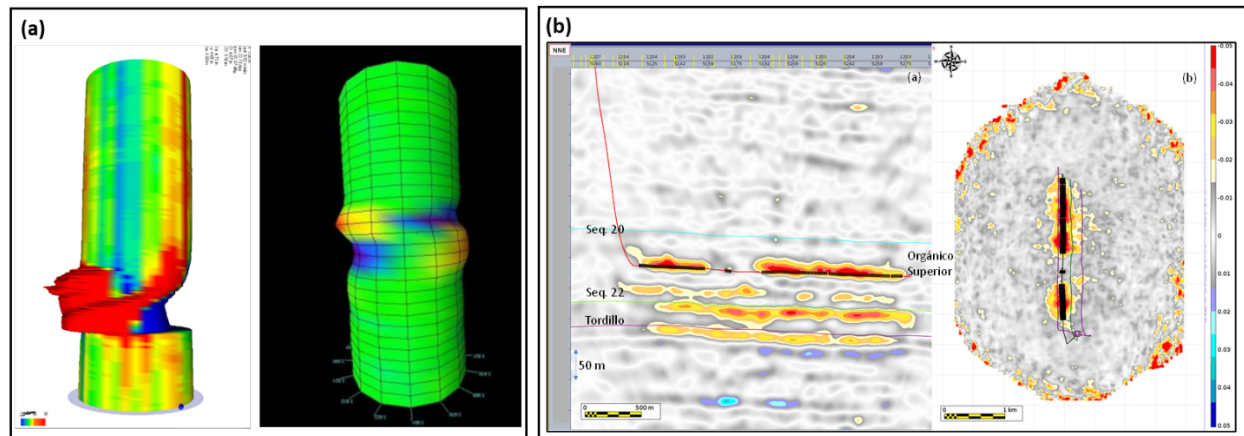


Figure 1: (a) Casing shearing in vertical wells. (b) 4D Seismic in well with casing restriction

This type of restrictions has been identified anywhere along the lateral. In general, when encountering issues in a well during completion, it is crucial to identify the location of the restriction or the affected area in

terms of meters of the drain. For instances occurring along the horizontal section (Toe or middle section) or involving a small area, operators often opt for implementing a contingency plan in the affected zone. This alternative involves the utilization of Slim Bottom Hole Assembly (BHA) guns and dissolvable plugs, along with modifications in stage length and fracture intensity (both fluid and proppant). The primary objective in these scenarios is to maintain the regular zipper sequences and preserve the original front of the completion. Another approach is to modify the zipper sequences to minimize the impact, utilizing the knowledge and understanding of the problem's root cause.

A more complex situation was observed in wells with problem at the heel section, this implies a risk of loss the entire drain section. In such cases, some operators have chosen to reschedule the intervention. For these wells, a contingency completion was prepared, considering various factors as following.

- *Acquisition*: obtaining information regarding the type, form, dimension of the restriction, and the accessibility of the lateral section is essential. This enables to determine the length of the tool-string or select the appropriate perforation system to be employed - Wireline Gun (WLG) or Abrasive with Coiled Tubing (CT).
- *Completion design*: it is necessary to determine the desired length of the stimulated section during the extended stages, as well as the number of stages in the well (Figure 2a). The type of isolation, such as dissolvable plugs or diverters, should also be determined. Additionally, the number of perforations per stage, the application of diverters, the fracture intensity in terms of fluid and proppant, the fracture rate, and the operation time for the extended stages need to be considered.
- *Logistics*: the completion of this type of contingency requires a considerable amount of equipment and materials to be present at the location within a short timeframe. These include water, chemicals, proppants, fuel, equipment, maintenance, and human resources (Figure 2b). Additionally, specialized tools such as the slim BHA and plugs, as well as coiled tubing, must be readily available (Figure 2c). In VM wells, examples of this completion technique have shown the use of 4 to 6 extended stages (with 3 internal sequences), involving 16 to 20 hours of continuous pumping, with a total of 2,050,000 pounds of proppant and 40,000 barrels of fluid.
- *Economy*: the casing restrictions significantly affect the cost of the well. In many cases, this contingency leads to a compromised design to stay within the assigned budget and minimize the additional costs caused by this situation.

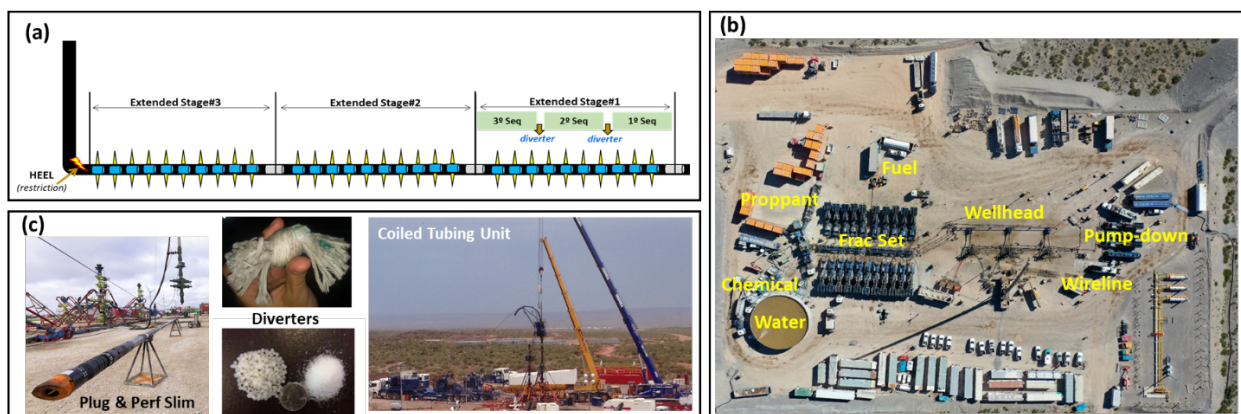


Figure 2: (a) Completion design - extended stages. (b) Logistic and resources at well location. (c) special equipment and tools

In general term, there has been some observation after applying these treatments: a) from an operational standpoint, this type of completion has been effectively implemented; b) usually, these wells are underperforming when compared to wells with typical design (intensity) and no issues, and c) adverse effects on nearby wells resulting in fracture-driven interactions (FDI) and production loss. Some of the possible explanation about these results could be the compromise design (lower intensity) is not the appropriated, a poor diversion in extended stages (under or over stimulated zone) contribute to well performance and the delay in completion can generate areas more prone for FDI event and subsequent implications.

Considering the previous point mentioned is necessary to improve the approach for this kind of contingency completion to obtain better results and avoid generate collateral negative impact.

History of casing restriction in Fortin de Piedra Field

The Fortin de Piedra (FP) block is in the central part of the Neuquén Basin with a surface area of 248 km² that is evenly distributed within the wet gas and dry gas windows (Figure 3a). The exploration phase consisted in 1 workover and 2 vertical wells (2013-2016), then an appraisal phase was completed with 5 horizontal well (2016-2017) to finally enter at the development (2017 up to date) with 126 horizontal wells. A more detailed information about geology, reservoir and other aspect of the field has been presented in Biscayart et al. (2020).

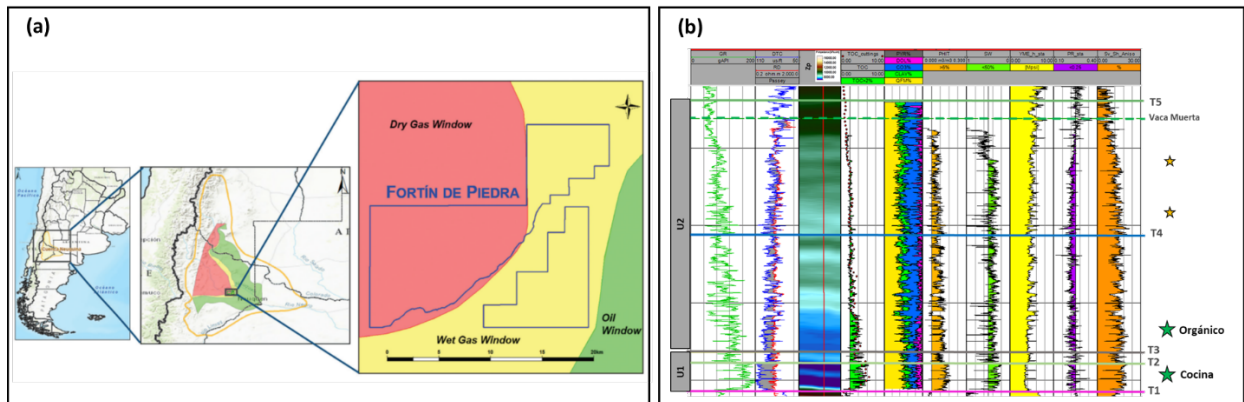


Figure 3: (a) Location map of Fortin de Piedra (Neuquén Basin – Argentina). (b) Reservoir characteristic and target zone in Vaca Muerta

From the beginning, casing restriction in FP has not been an important issue, a total of 131 horizontal wells and 5300 fracture stages were executed covering the entire field. A detailed review in completion database (wireline, fracture, and coiled tubing operations) was performed to validate wells which suffered this problem, focusing in three main aspects that impact negatively.

- *WL BHA stuck events (fishing)* - Along the activity developed, several situations with BHA were detected. The standard procedure in case with a BHA stuck were applied (maneuver to release, setting the plug, perforate cluster, release WG). No fishing has been observed during these years related to casing restrictions as main cause.
- *Unstimulated drain (meters)* - A total of 25 wells were affected (19% of the well at the field); this problem generated 2,078 meters of unstimulated drain, representing less than 1% of the total meter planned to be completed (306,700m). Most of the wells shown 1 or 2 event (less than 5% of drain in meters), the worst case observed consisted in 8 CR representing a loss of 320 meter (13% of

drain). The percentage of occurrence by target zone has been similar, 17% in wells located at “cocina” and 20% at “orgánico”. (Figure 3b)

- CT-Post Fracturing Cleanup (% of drain clean) - All the wells were fully cleaned the lateral section (97 to 100%), most of the case using an only size mill, in some cases was necessary to change and reduce the size to pass the restriction. The following is the information for mill depending on the casing size. In 4 ½-inch casing the standard mill was 85mm (82 to 89mm) and in 5-inch casing the standard mill was 87mm (85 to 92mm).

More recently, (April 2023) an unusual condition was detected during the beginning of 3-well PAD “A” completion, two of them at the first run of plug and perforation (PnP) presented a sudden stop with drop in surface tension. After repeated maneuvers at the MSP (maximum safe pull), it was decided to release the tool. After that, CT was run in order to fish the BHA, calibrate the well and perform a multifinger imaging log in order to obtain information along the lateral.

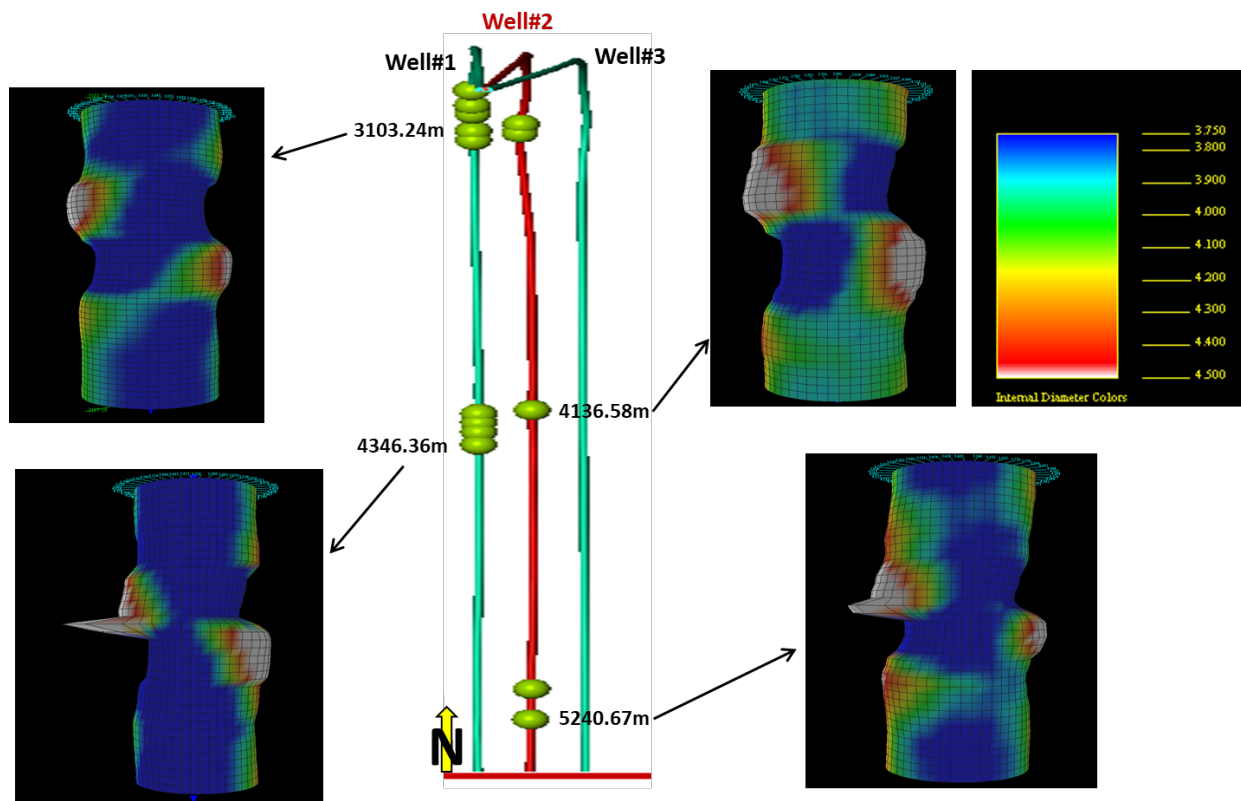


Figure 4. Multifinger Imaging Tool, main deformations detected

Anomaly Detector Tool (ADT) – Initial Concept

Abnormal casing collar locator readings can offer insights into potential wireline obstructions during subsequent runs. Although not specifically intended for detecting casing restrictions, these atypical CCL variations may indicate discrepancies in casing diameters.

During our pursuit of examining variables in cases involving wireline challenges, we found that most instances exhibited CCL irregularities appearing at multiple stages before issues arose. Consequently, we delved further into the realm of CCL parameters and develop an Anomaly Detector Tool (ADT).

The high-severity events experienced in PAD “A” make us prioritized the development, since the ADT's initial tests demonstrated its capabilities in identifying precursory warning signs well before complications occurred.

Anomaly Detector Tool (ADT) – Model Basics

Anomaly detection techniques play a crucial role in various domains, but many existing methods encounter limitations in terms of detection performance or computational complexity.

We develop a machine learning algorithm to identify anomalies using an anomaly score (s). Instances with scores nearing ~ 1 are deemed definite anomalies, whereas those with scores below 0.6 are classified as normal instances.

The Anomaly Detection Tool (ADT) identifies all inconsistencies within the data, such as casing collars (CC) and potential casing restrictions (CR). With knowledge of the CC position gathered from the casing run tally, it then removes the CC from the relevant data. Subsequently, an alarm graph is created, displaying potential CRs alongside their associated anomaly scores.

This model efficiently operates with partial models and small sample sizes, eliminating the necessity for complete isolation of every standard instance. Consequently, it serves as a resource-efficient and effective solution for detecting anomalies in wireline runs.

Anomaly Detector Tool (ADT) – Model Normalization

In the context of dealing with varying minimum and maximum values in CCL, we employed data normalization as a solution. Factors such as wireline velocity can influence the CCL value, making it diverse. To standardize these values, we applied a scaling factor of 10 during normalization. This process involved multiplying the CCL values by 10 and subsequently dividing them by the highest CCL value detected.

Anomaly Detector Tool (ADT) – Sampling Frequency

We initially employed a 1-second real-time CCL sampling. However, this approach resulted in a higher than expected rate of overlooked anomalies, with even CC being overlooked. To address this issue and to optimize the use of the data, we initiated a more comprehensive examination of the sampling rate.

Our approach involved liaising with the wireline company to obtain a Log ASCII Standard (.las) file of their ongoing tests. This crucial resource enabled us to experiment with different sampling intervals, thereby facilitating the identification of the best-suited frequency for our needs.

Our investigation revealed that a STEP (sampling distance) of 0.01m was necessary to effectively identify all anomalies during test sampling.

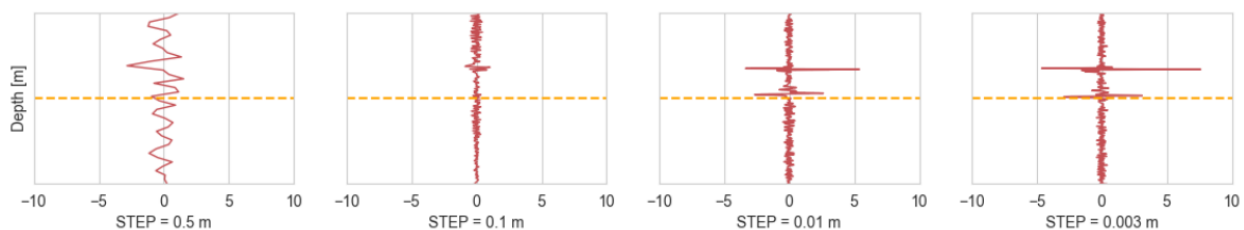


Figure 5. CCL sampling frequency examples

Figure 5 illustrates a scenario where there is no observable CR with the ADT in STEP 0.5m and 0.1m. Complementary, Figure 6 evaluates the steps of 0.5m (represented in red), 0.1m (black), 0.01m (blue), and 0.003m (green) through a series of histograms. It is noteworthy that only the 0.01m and 0.003m steps exhibit a distribution of anomalies, indicative of potential CRs, as demonstrated in Figure 6.

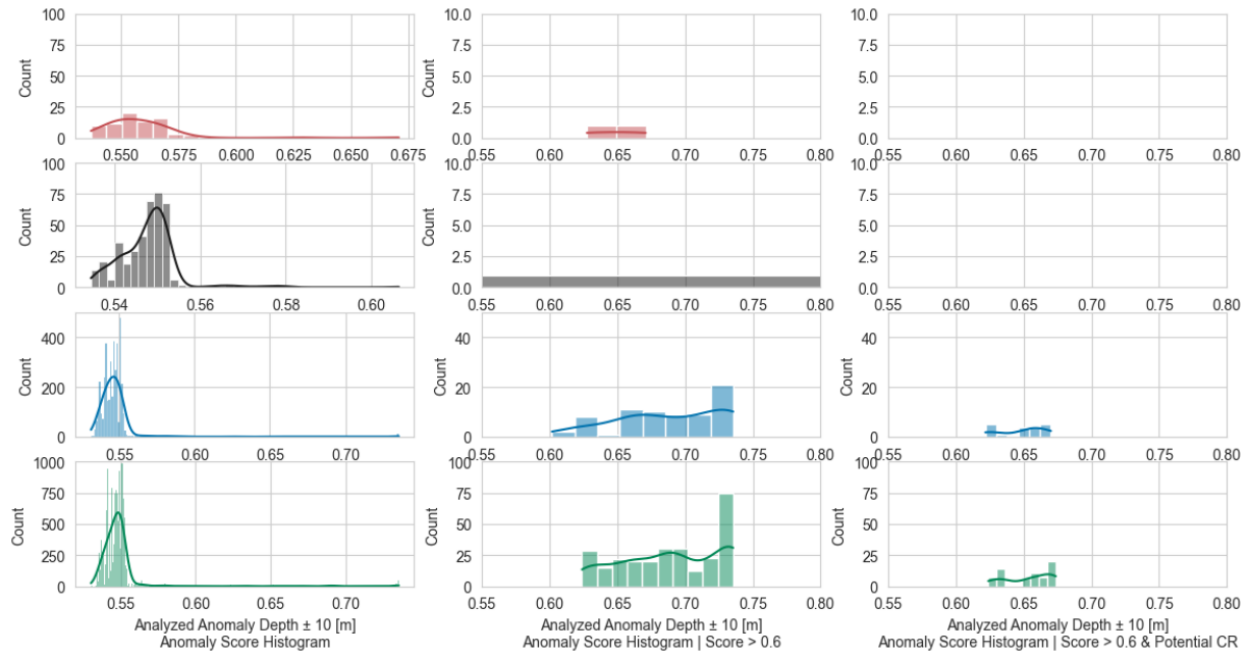


Figure 6. Anomaly Score Histogram: evaluation of various sample steps: 0.5m (represented in red), 0.1m (in black), 0.01m (in blue), and 0.003m (in green). The first graph shows the distribution of all anomaly scores. The second graph focuses on scores greater than 0.6. The final graph highlights potential CR scores.

We found that a STEP of 0.003m did not provide additional information and necessitated extra processing in our actual model. As a result, we concluded that a STEP of 0.01m is preferable when using ADT.

Anomaly Detector Tool (ADT) – Sampling Size

The challenge of swamping - a phenomenon characterized by normal instances being in close proximity to anomalies, thereby increasing the number of partitions required to segregate anomalies - is addressed in our work. One key strength of our algorithm lies in its ability to identify non-conforming samples without necessitating the construction of a data profile. This is a distinct departure from several existing methods that demand large samples. Our model continues to demonstrate robust performance even when sample sizes are maintained at a minimal level.

In the context of identifying anomalies during a wireline run, our model is employed every 50 meters in a loop. Prepared to handle real-time data, the model does not require large data quantities. This makes it a practical and effective tool in real-time scenarios.

Anomaly Detector Tool (ADT) – Tuned

Utilizing the multi-finger caliper tool in the PAD “A” allowed us to fine-tune the ADT. The calibration of the ADT through a multi-finger run demonstrated empirical evidence linking casing deformations with additional readings.

This adjustment employed data derived from four wireline runs implemented on the problematic well that we had the multifinger information. Figure 7 depicts the multifinger plot related to the ADT figure with CCL data in one point of interest.

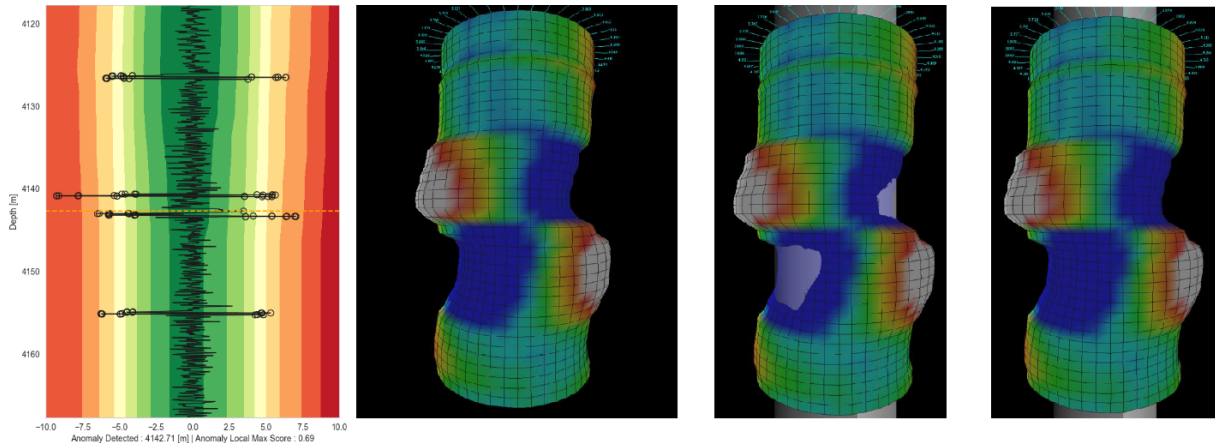


Figure 7. ADT Anomaly detected vs Multifinger caliper simulation.

Anomaly Detector Tool (ADT) – Blind Tests

After the Anomaly Detection Tool (ADT) was calibrated, we conducted blind tests using wells known for their problematic conditions. Figure 8 displays an instance where the wireline was obstructed during the 22nd run.

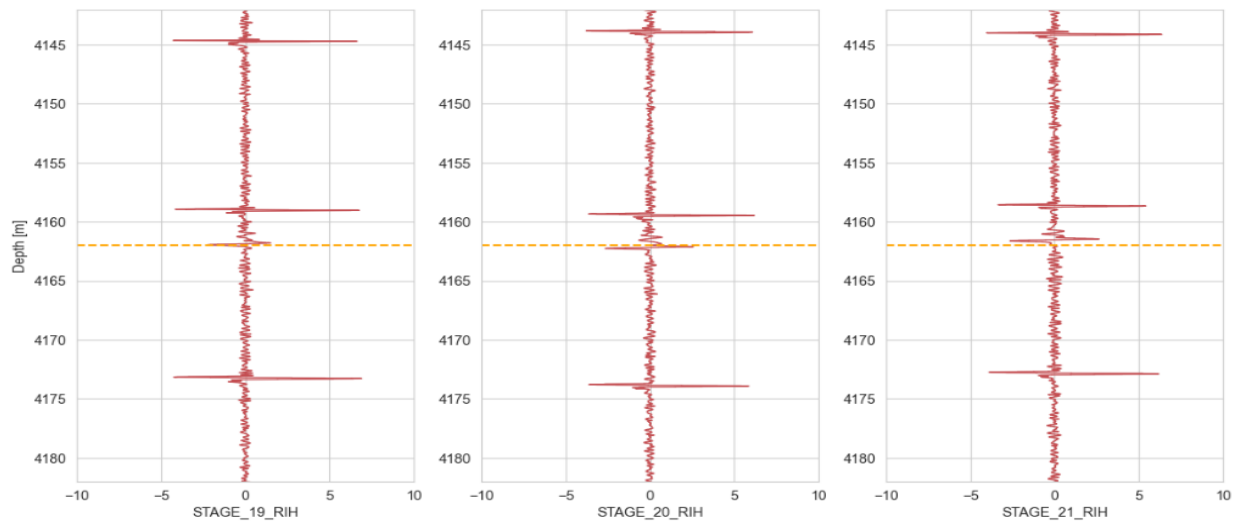


Figure 8. CCL consecutive runs, raw data, STEP 0.01m.

Remarkably, the ADT was capable of detecting the anomaly several stages earlier than expected, as visually demonstrated in Figure 9.

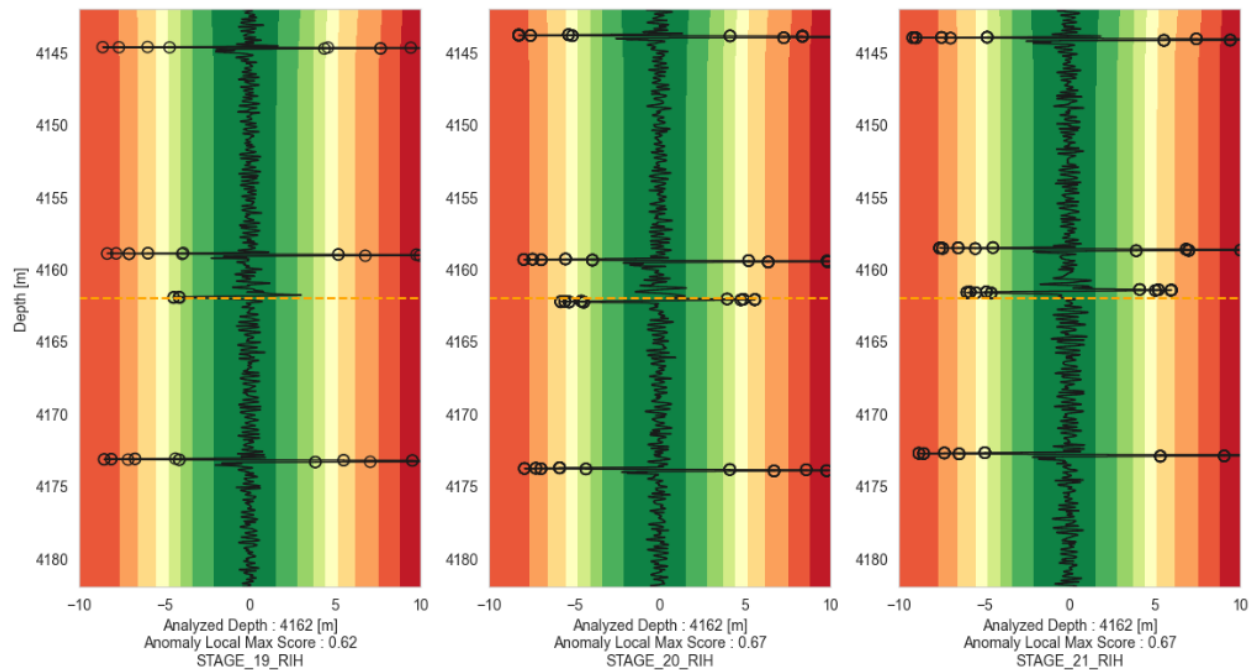


Figure 9. ADT Anomaly output.

Conclusions

Casing deformations in Vaca Muerta can result in serious issues like the loss of unstimulated drain and reduced productivity. It becomes essential to design alternative completions for wells with casing restrictions, which can cause operational complications and adverse economic consequences.

While casing deformations have not traditionally been a significant concern in Fortín de Piedra, recent observations of deformations in a PAD "A" with two horizontal wells have prompted the development of the ADT tool for quick and early detection. ADT was tuned with the well in PAD "A", and has successfully identified anomalies in blind tests.

To establish a connection between casing deformations and the likelihood of success with specific plugs or BHA, we need more statistical data. Furthermore, in later work phases, we are exploring real-time data reading technologies for future pads, which will enable us to detect anomalies and establish a contingency plan for an effective response.

References

Biscayart, P., Brolli, M., d'Hiriart, J., et al 2020. "Factory Mode" Development of Fortín de Piedra Block, Gas Window, in Daniel Minisini, Manuel Fantin, Ivan Lanusse Noguera, and Hector Leanza eds. Integrated geology of unconventional: The case of the Vaca Muerta play, Argentina: AAPG Memoir 121, p. 515-528
[DOI: 10.1306/13682239M1203843](https://doi.org/10.1306/13682239M1203843)

Bonapace, J.C., Riolfo, L., Zrain, R., et al 2023. From Exploration To Development: The Completion Evolution In Vaca Muerta. Presented at the SPE Argentina Exploration and Production of Unconventional

Resources Symposium, Buenos Aires, Argentina, 20–22 March. SPE-212574-MS <https://doi.org/10.2118/212574-MS>

Casero, A. and M. Rylance 2020. The Unconventional Unconventionals: Tectonically Influenced Regions, Stress States and Casing Failures. Presented at the SPE Hydraulic Fracturing Technology Conference and Exhibition, The Woodlands, Texas, USA, 4-6 February. SPE-199710-MS <https://doi.org/10.2118/199710-MS>

Chao, G., Canatelli, A., Sainz-Trapaga, M., et al 2022. Caracterización de volumen de roca estimulada (SRV) en un PAD Multi-Well Multi-Landing en Vaca Muerta. Presented at the IAPG XI Congreso de Exploración y Desarrollo de Hidrocarburos, Argentina, Mendoza, 8-11 November

Gandomkar, A., Katz, D., Gomez R., et al 2021. Geomechanically Driven Casing Deformation During Multistage Perf and Plug Fracturing Operations: Investigation and Mitigation. Presented at the SPE Europec featured at 82nd EAGE Conference and Exhibition, Amsterdam, The Netherlands, 18-21. SPE-205221-MS <https://doi.org/10.2118/205221-MS>

Garcia, N., Sorenson, F., Bonapace, J.C., et al 2013. Vaca Muerta Shale Reservoir Characterization and Description: The Starting Point for Development of a Shale Play with Very Good Possibilities for a Successful Project. Presented at the SPE Unconventional Resources Technology Conference, Denver, Colorado, USA, 12-14 August. URTEC-2013-090-MS <https://doi.org/10.1190/urtec2013-090>

Maria, S., Rodriguez, L., Herrera, H., et al 2022. Evolución de un Proyecto No Convencional Greenfield en la Fm. Vaca Muerta, Yacimiento La Calera, Cuenca Neuquina. Presented at the IAPG XI Congreso de Exploración y Desarrollo de Hidrocarburos, Argentina, Mendoza, 8-11 November

Mirande, J.M, Palombizio, G., Arevalo, L., et al 2023. Novel Coiled Tubing Post-Fracturing Cleanout of Unconventional Wells Without Downhole Motor: A Case Study from Argentina. Presented at the SPE/ICoTA Well Intervention Conference and Exhibition, The Woodland, Texas, USA, 21-23 March. SPE-212943-MS <https://doi.org/10.2118/212943-MS>

Rimedio, M., Shannon, C., Monti, L., et al 2015. Interference Behavior Analysis in Vaca Muerta Shale Oil Development, Loma Campana Field, Argentina. Presented at the SPE Unconventional Resources Technology Conference, San Antonio, Texas, USA, 20–22 July. URTEC-2154850-MS <https://doi.org/10.15530/URTEC-2015-2154850>

Russel, T. 2021. Well Shear Associated with Conventional and Unconventional Operations: Diagnosis and Mechanisms. SPE Prod & Oper 36 (2021): 427–444. SPE-205007-PA <https://doi.org/10.2118/205007-PA>